ARKA JAIN University Jharkhand			Question Bank
Branch	**Computer Science & Engineering**	Program	**BTECH**
Course Name	**COMPUTER ORGANIZATION & ARCHITECTURE**	Semester	**IV**

Section A: [2 marks each]

1. Explain the types of memory.
2. What is main memory?
3. What is auxiliary memory?
4. Explain about Arithmetic and Logic Unit.
5. Define ALU, control unit, input and output unit.
6. Define single bus and multiple bus structure.
7. Explain about central processing unit
8. Define opcode and offset.
9. Define MAR and PC.
10. Define GPR and IR.
11. Differentiate between the static and dynamic RAMs.
12. What is pipelining?
13. Write advantages and disadvantages of pipelining.
14. What is cache?
15. What is the need for using cache memory?
16. Define cache hit and cache miss.
17. What is cache coherence problem?
18. What is pipeline bubble?
19. Describe addressing Modes.
20. Explain any two addressing modes with the help of examples.
21. What is cache memory mapping? List the types of mapping policies used.
22. Perform the arithmetic operations (-32) + (-11) in binary using signed 2's complement representation for negative numbers.
23. Perform the arithmetic operations (-32) – (-11) in binary using signed 2's complement representation for negative numbers.
24. Perform the arithmetic operations (-25) + (-21) in binary using signed 2's complement representation for negative numbers.
25. Perform the arithmetic operations (-65) – (-33) in binary using signed 2's complement representation for negative numbers.
26. List the steps involved in an instruction cycle.
27. Explain the importance of different addressing modes in computer architecture.
28. What is hit-ratio?
29. What are the classifications of 8086 interrupts?
30. What is synchronous data transfer?
31. What is asynchronous data transfer?

Section B: [5 marks each]

32. Explain address bus, data bus and control bus.
33. Discuss bus structure and why it is used?
34. Explain the functional units of a computer.
35. Compare and contrast RISC and CISC architecture
36. What are pipelining hazards? What are their types?
37. Explain the functional units of a computer.
38. Discuss the various types of cache miss.
39. Explain Write-through and Write back cache write method.
40. Write down various addressing modes used in microprocessors. Explain any three of them with examples.
41. Discuss various flags available in status register of 8086.
42. Discuss the steps involved in program execution.
43. What are pipelining hazards? What are their types?
44. Explain in detail about the instruction cycle.
45. Describe different types of interrupts in a microprocessor system
46. Explain Write-through and Write back cache write method.
47. Write down various addressing modes present in 8086.
48. A non-pipelined system takes 50 ns to process a task. The same task can be processed in 6 segment pipeline with a clock cycle of 10 ns. Determine approximately the speed up ratio of the pipeline for 500 tasks.
49. Assume that for a certain processor, a read request takes 50 ns on a cache miss and 5 ns on a cache hit. Suppose while running a program it was observed that 80% of the processor's read request result in a cache hit. Find out the average read access time.
50. A cache memory needs an access time of 50 ns and main memory needs an access time of 150 ns. What is the average access time of CPU? Assume hit ratio = 80 %.
51. Represent the following numbers in 2's complement form:

a. 32	e. 10
b. -32	f. -10
c. 9	g. 63
d. -9	h. -63

Section C: [10 marks each]

52. What do you mean by addressing mode? Explain any five addressing modes in detail.
53. Describe pipelining with example and diagram. Write its advantages and disadvantages. Discuss various pipeline hazards as well.
54. Draw and explain functional block diagram of 8086 microprocessor.
55. What is the need to implement memory as a hierarchy? Elaborate in detail the memory hierarchy with neat diagram.
56. Explain Flynn's taxonomy in detail.
57. Perform below multiplication using booth's multiplication algorithm for signed numbers: 9 x (-10)
58. Perform below multiplication using booth's multiplication algorithm for signed numbers: (-9) x 10

59. Perform below multiplication using booth's multiplication algorithm for signed numbers: 9 x 10
60. Perform below multiplication using booth's multiplication algorithm for signed numbers: 3 x (-7)
61. What is cache memory mapping? Explain Associative and Set Associative memory mapping.
62. Explain Direct memory mapping technique.
63. Explain cache coherence problem and various software and hardware solutions for it.
64. Express in detail about Carry look ahead Adder.